Determinants of Attendance in the English Women's Super League: Does Demand for Men's Football Spillover? *

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Abstract

Interest in women's sports, particularly football, has surged in recent years, with growing spectatorship and media coverage. Despite this, women's football in England remains in an early stage of growth, with attendances significantly below men's football. This study examines the determinants of attendance in the English Women's Super League (WSL) from 2011 to 2023, focusing on potential spillover effects from men's football. Analysing 775 matches, we review how the men's team performance and brand strength influence attendance at women's matches. It is found that three main factors drive WSL attendance: (1) team and game characteristics, such as team performance, weather, and team rivalry; (2) success of the national team; and (3) performance and characteristics of the men's team, showing a significant spillover effect. Such results offer key insights to stakeholders, as considering spillover effects can enhance investment strategies and foster growth in women's football.
Keywords: *Demand Spillover in Sports; Demand Analysis; Brand Integration; Competitive Balance.*
JEL Classification: *Z2; L2; M3.*

1 Introduction

Interest in women's sports, particularly football, has surged in recent years, with record-breaking audiences and increased media coverage, aligning with FIFA's objective of making the growth of women's football a top priority.[1] In particular, the English Women's Super League (WSL) has experienced a sharp increase in average stadium attendance since the commencement of the league in 2011, from 550 in its inaugural season to 5,387 in 2022/23. However, consumer interest in the WSL is still far from its male counterpart. For example, the average attendance for Chelsea F.C. Women in the 2022/23 season, when the side won the league, was around 6,000. In comparison, the men's side saw an average attendance of around 40,000 in the same season despite disappointing performance.[2] A peculiar feature of the WSL is the association of women's teams with football clubs traditionally focused on the men's team, with all teams in the 2022/23 season of the WSL being integrated with football club's playing in the men's professional leagues. Such team integration provides a range of benefits for women's teams, especially in their early stages of growth, such as improved training facilities or greater financial resources (Valenti, 2019; Welford, 2018). Similarly, women's teams may benefit from the brand spillover of as-

*We thank David Forrest, Babatunde Buraimo, and Kaori Narita for helpful comments on an earlier draft of the paper. We also extend our thanks to Professor Mario Lackner and the two anonymous referees for their suggestions on a past version of this paper.

[1] See fifa.com/womens-football.

[2] See economist.com

sociation to a large football club (Valenti, Scelles, & Morrow, 2020). The present paper aims to disentangle such brand spillover effects on the demand for English women's football teams, analysing whether the success of the men's team increases attendance at WSL matches. We expand on this by reviewing the general determinants of attendance within the WSL, along with testing the effect that success by the English women's national team in international competitions has on the demand for domestic football.

While sports attendance analysis is a well-established field within the sports economics literature, the analysis of attendance demand for women's football has only recently received considerable attention (see, for example, Hadwiger, Schmidt, and Schreyer (2024), Meier, Konjer, and Leinwather (2016), and Valenti et al. (2020) for a handful of great recent studies covering this topic). Our contribution to the sport economics literature, as such, is twofold. Our most general contribution is to provide a set of determinants of attendance for matches within the top tier of English football from the league's commencement in 2011 to the 2022/23 season. In doing so, we can understand how consumer interest is shaped and review if the demand for football is shared across men's and women's football. Indeed, it may be assumed that the demand for, and product of, the football teams is similar; however, findings by Agha and Berri (2023) and Konjer, Meier, and Wedeking (2017) suggest this may not necessarily be the case, finding that the product and the demand functions differ between men's and women's basketball and tennis. By modelling the general determinants of demand and comparing our results to the broader attendance demand literature, we can uncover the extent the demand functions are similar. Our second contribution is more specific. We analyse the demand spillovers from the association and success of men's football teams and the English women's national team. As noted, the association of women's football clubs create unique avenues for brand strategy through the association with big-name football clubs that were previously oriented around the men's team. Due to this, understanding whether interest in women's football can benefit from outside sources is of key relevance for stakeholders in the football industry.

Using match-level data from 775 matches, this paper models attendance demand for the WSL from the 2011 to the 2022/23 season. The specification includes a rich set of controls capturing team and match characteristics, as well as performance and economic indicators. We further control for the fixed effects of the home and away team, season, and day and month. To preview, we find that WSL attendance is determined by many traditional variables employed in modelling of attendance at men's football, such as team performance, weather conditions, rivalry, geographical distance between the home and away clubs, and the ex-ante probability of home victory. We also include a measure of outcome uncertainty and, consistently with much of the literature on men's football, find outcome uncertainty does not significantly explain variations in stadium attendance. Second, we find an increase in attendance over the analysis period. Such growth in consumer interest is especially evident after the summer of 2022, when the English women's team won the Women's European Championships. In line with Meier et al. (2016) and Stephenson (2024), this suggests a strong connection between consumer interest and national team performance. Our third result is that the previous performance of the men's team predicts attendance at the women's team even after controlling for home and away team fixed effects. This indicates the existence of an overlap of fans committed to the brand and willing to support the women's team. The main results stand up to different robustness exercises consisting of including match significance indicators, lagged attendance, and fixture-specific dummy variables. This paper proceeds as follows. The following section (2) positions the paper in the literature and states its contribution. Section 3 describes the main characteristics of the WSL. The dataset employed in the analysis is presented in Section 4. We conduct the empirical analysis and test the robustness of the model to different considerations in Sections 5 and 6, respectively. Finally, Section 7 discusses the implications of the results and concludes.

2 Related Literature

The question of why spectators turn out to the stadium to watch a sporting event is of key interest to scholars and is fundamental to better understanding sport business. The framework for a sports consumer demand function was introduced in the seminal work by Rottenberg (1956) who outlines attendance as a function of income, price of admission relative to substitutes, and competitive balance; a positive function of a team's market size, the size and location of the team's stadium, and average league position; and a negative function of the quality of substitutes. Since Rottenberg (1956), a plurality of papers within the sports economics and management literature have provided econometric analyses of spectator demand, typically modelling attendance as a function of various product-specific and socioeconomic factors. A review by Borland and MacDonald (2003) suggests there are five categories that model the determinants of demand – Economic factors (e.g. price, opportunity cost of attendance, and macroeconomic factors), viewing quality (e.g. timing of the match and weather), characteristics of the sporting event (e.g. match outcome uncertainty and team quality), consumer preferences (e.g. habit persistence), and stadium capacity.

Despite the academic interest in stadium demand, most of the focus has remained on men's sports. A recent review by Schreyer and Ansari (2022) notes the small size of research on the demand for women's sport, highlighting only three papers (out of the 195 reviewed) that touched on stadium attendance demand. However, since 2022 there has been a considerable expansion in demand literature regarding women's sport, with multiple manuscripts covering the demand for basketball (Agha & Berri, 2023), tennis (Konjer et al., 2017), volleyball (Baecker, Chan, Schmidt, Schreyer, & Torgler, 2023), and football (Chahardovali, Watanabe, & Dastrup, 2023; Hadwiger et al., 2024; LeFeuvre, Stephenson, & Walcott, 2013; Meier et al., 2016; Meier & Leinwather, 2012; Stephenson, 2024; Valenti et al., 2020; Valenti, Scelles, & Morrow, 2024). Despite the increase in the breadth of the literature, the number of studies, relative to those covering male football, remains low.

Although this may be understandable if the consumption of men's and women's sports shared similar characteristics and reacted to the same inputs, studies have pointed out that such determinants are (at least partially) different, suggesting the product may not be completely similar. For example, Agha and Berri (2023) find that the determinants of stadium attendance differ both in terms of size and significance between elite women's and men's basketball. Konjer et al. (2017) find similar results when comparing television spectatorship for men's and women's tennis.

As noted, there exists a handful of studies that review stadium attendance demand in women's football, covering American (Chahardovali et al., 2023; LeFeuvre et al., 2013; Stephenson, 2024), English (Valenti et al., 2024), German (Hadwiger et al., 2024; Meier et al., 2016), French and Swedish (Hadwiger et al., 2024), and Pan-European (Valenti et al., 2020) competitions. It is found that the demand for women's football reacts similarly to men's football and across nations. Sport-specific elements, such as the consumer preferences and match-specific characteristics, have shown to be important determinants for stadium attendance. Notably, demand is a positive function of the quality of the competing teams (LeFeuvre et al., 2013; Meier et al., 2016; Stephenson, 2024; Valenti et al., 2024), along with the presence of superstars (LeFeuvre et al., 2013; Stephenson, 2024). Similarly, demand for women's football shows elements of habit persistence (Meier et al., 2016).

Specific matches within a season are also shown to increase attendance, particularly, inaugural matches (LeFeuvre et al., 2013; Stephenson, 2024), matches played late into cup competitions (Valenti et al., 2020), along with 'double-header' matches (LeFeuvre et al., 2013; Stephenson, 2024; Valenti et al., 2024). Peculiarly there are mixed results on the effect competitive balance has on attendance, with Meier et al. (2016) and Stephenson (2024) rejecting the uncertainty of outcome hypothesis, like findings on male football (Baydina, Parshakov, & Zavertiaeva, 2021). However, Valenti et al. (2020, 2024) finds competitive balance has a significant, but conflicting, effect on women's football attendance, with increased outcome uncertainty increasing demand in European cup competitions (Valenti et al., 2020) but decreasing spectatorship in England (Valenti et al., 2024).

The viewing quality of the football match show varying effects. Adverse weather is a negative function of demand with matches with increased precipitation seeing lower attendance (LeFeuvre et al., 2013; Meier et al., 2016; Stephenson, 2024; Valenti et al., 2024). Similarly, matches played in Europe see higher demand when the temperature is higher (Hadwiger et al., 2024; Meier et al., 2016; Valenti et al., 2020, 2024). Like the other viewing quality characteristics, the stadium the football match takes place in has mixed effects. Meier et al. (2016) were the first to review how demand changes with the quality of the stadium, finding no statistically significant effect; however, this effect appears to differ in other leagues. Valenti et al. (2024), studying English football, finds the opposite effect, with attendance increasing with the size of the stadium. Somewhat similarly,Chahardovali et al. (2023) find that there is a drop off in demand for matches played in stadiums that are further from the team's city centre. Finally, Hadwiger et al. (2024) and Stephenson (2024)find that matches played in the women's teams associated men's team stadium see large increases in attendance. These results suggest that women's football consumers do respond to the quality of the stadium the match takes place in; or, that teams may benefit from association to the men's team by utilising facilities such as the stadium. We follow this analysis by reviewing how playing in the men's home stadium as a special relocation or normal match affects demand, this will be touched upon in the empirical framework section.

Like our study, other papers have reviewed potential spillover effects on demand from success from national team's and affiliation with the men's football team. Regarding national team spillover effect, LeFeuvre et al. (2013) and Stephenson (2024) find that American attendances increase in the season following the women's world cup, with Meier et al. (2016) finding a similar result in Germany but noting that the effects are short-lived. Potential spillover effects from male football teams have been analysed by reviewing the effect of integration with a male team under one football club, however, it has been found there is little effect with Meier et al. (2016) finding that teams that switch to being an integrated club don't see increased attendance, and Valenti et al. (2020) finding that demand is only affected when the away team is integrated and plays within a big-5 league. Valenti et al. (2024) take a slightly different approach to measuring male team spillover effects by using the average attendance of the male football team to measure the popularity of the brand but find insignificant effects on demand. Despite these, Hadwiger et al. (2024) suggest that women's teams do see a demand benefit through integration with the men's team, however, this comes from effective management of the women's team, as integration itself may harm demand.

Our study looks to expand on this literature to study the spillover effects on demand within the WSL. Firstly, we analyse the effect that national team success has on attendance to domestic club football. Secondly, we expand on the approaches by previous studies by reviewing how different measures of male team quality and success (as a proxy for the quality of the brand) effects the demand for the associated women's team.

3 English Women's Football and the Women's Super League

The history of English women's football is storied but stained with controversy. Despite the perception that female participation in football is a recent phenomenon, the history of women's football goes back centuries. In the early 1900s, popularity over the women's game gained momentum following the first world war up to a peak in December 1921, where a match between Dick Kerr, Ladies and St. Helens Ladies drew a crowd of 53,000 at Goodison Park (Tate, 2013). To spite the popularity of women's football, the Football Association (FA) would ban women from playing on association pitches in the same year, with the ban remaining for 50 years. Although women's teams would continue to play in smaller rugby venues, the women's game would be quickly overshadowed by men's team football, struggling to gain a large foothold in English football since.

To support the growth of women's football into the 2010s, the FA restructured the Women's Premier League National Division – the previous top tier of English Women's football – into the formation of the WSL. The WSL initially began as a summer league, running from March to October. This format was set with the intention to make the league as attractive as possible from a broadcasting and commercial standpoint, avoiding scheduling conflicts with male football and allowing for the development of a 'niche' product (The Football Association, n.d.). However, the format was scrapped and the league restructured into a winter league in 2017, due to concerns over player welfare and as a strategy to develop the league and improve attendance (The Football Association, 2016).

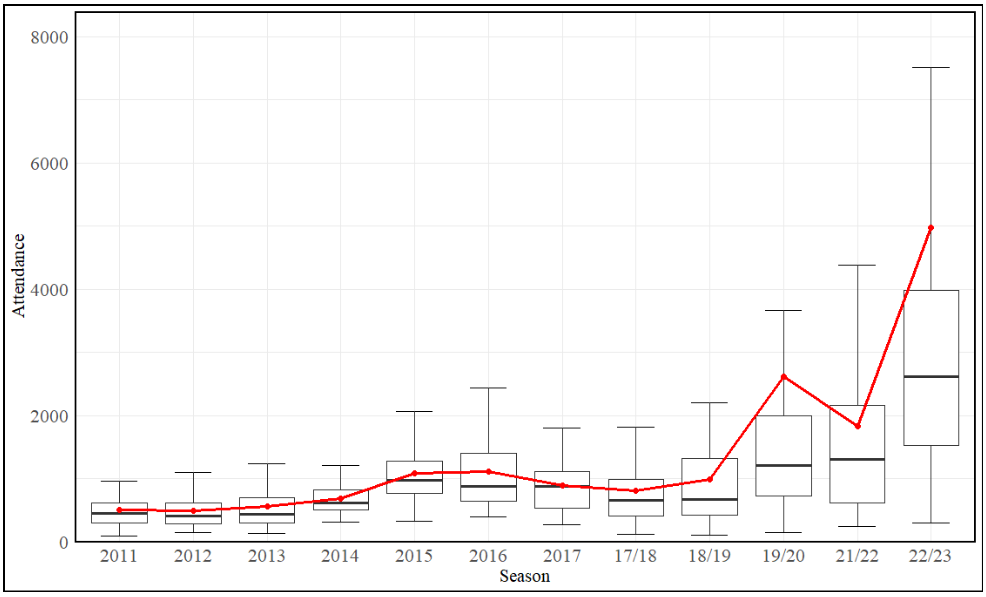
During the commencement of the WSL, admission to the league was granted following successful two-year licence applications for 8 teams. To be granted a license, teams had to meet the FA's financial requirements, including paying their top four players £20,000–£30,000 and matching the FA's annual investment of £70,000 for the first two years (Leighton, 2009). Additionally, teams were required to fulfil commercial, facilities, and staffing criteria. The licencing criteria by the FA reflected their strategy for establishing a financially viable and successful semi-professional women's league. However, this approach proved controversial, as smaller clubs that without ties to larger English League teams struggling to meet such requirements.[3] Following the licencing process, eight teams formed the inaugural season of the WSL. Five of these teams were affiliated with Premier League clubs, namely Arsenal, Birmingham City, Chelsea, Everton, and Liverpool. The other teams, Bristol Academy (now Bristol City W.F.C), Doncaster Belles (now Doncaster Rovers Belles L.F.C), and Lincoln Ladies F.C (rebranded as Notts County Ladies in 2014), were associated with clubs lower in the English league pyramid.

The FA would further restructure women's league football after the two-year start-up, forming the WSL 2 and distributing new licences. Again, this approach was controversial over the brand strategy that favoured teams with strong branding, as Doncaster Belles were demoted to the WSL 2 in place of Manchester City, whose men's team won the Premier League in 2014.

In furthering their efforts to expand the popularity of women's football, the FA set out "The Gameplan for Growth" in 2017, with three main goals set for 2020: consistent international team success, doubled female participation, and doubling the number of fans, which involved increasing average attendance from 1,047 to 2,020 (The Football Association, 2017). To achieve this, the FA set out several strategies, involving increasing the profile of the England Women's National Team and improving the commercial prospects of women's football in England. Following this, the FA restructured the WSL into a full-time and pro-

[3]For example, Leeds Carnegie, winners of the 2010 FA Women's Premier League Cup, withdrew their WSL licence application as they were unable to commit to the increases in the leagues proposed budgets (Leighton, 2010).

Figure 1: Trend in Attendance (2011 - 2022/23)



Note: The red points and line represent the mean attendance within each season; Outlier values have been hidden from the graph.

fessional league ahead of the 2018/19 season, setting out new licencing requirements that meant teams must provide a minimum of 16 contact hours a week for players, and an academy. These changes also increased the league's size to 11 (Garry, 2018). The league expanded further the following season, bringing the total number of teams to 12, which has remained at this number since.

The strategy set out by the FA in 2017 proved to be a success, with steady growth seen since the strategy's implementation. As Figure 1 shows, there has been substantial growth in average attendance since the formation of the league following the restructuring in 2018/19. Attendances have since grown to 5,387 for the 2022/23 season. One reason for this may be the continued media and fan attention received by the England women's national team, with the FA's Head of commercial and marketing stating that "the [2019] World Cup provided a backdrop for an unprecedented acceleration of the women's game" (The Football Association, 2020). Despite such growth, however, spectatorship still falls far short of that of male football, with the average attendances of the Premier League (40,236), EFL Championship (18,795), and EFL League One (10,691) proving significantly higher. However, attendance in the WSL for the 2022/23 season was like that of EFL League Two in the same season (5,712), showing the increasing foothold the competition has within English football.

The FA has continued its commitment to the growth of women's football in England, with the 2021 strategy highlighting its ambitious goals to maximise audiences and growth in commercial revenue and financial stability (The Football Association, 2021). Although the FA seem to be on target in meeting its average attendance goals, financial stability may still present issues for the future of women's football, with WSL clubs seeing an average loss of £1.4 million in 2019, and average debt up by 1,351% from 2011 to 2019, even with an increase in revenue of 590% (Clarkson & Philippou, 2023). Despite the financial struggles, the aggregate revenue for WSL clubs continues to grow, with the 50% increase in revenue from the 2021/22 season to the 2022/23 season being attributed to the improvement of matchday and commercial revenue following the 2022 Euros (Deloitte UK, 2024a). Although the average percentage of revenue coming from matchday income is not significantly different between women's and men's side (which is about 22% and 18% respectively (Deloitte UK, 2023, 2024b)), some women's teams receive a majority of their income through matchday revenue, such as Arsenal Women's at 58% (Deloitte UK, 2023). This highlights the importance of analysing what determines attendance for football clubs, which the present study aims to address.

4 Dataset and Variables

The present study analyses stadium attendance demand in the FA WSL from the inaugural 2011 season through to the 2022/23 season, covering twelve seasons and nineteen distinct women's teams. We omit the 2020/21 season from our analysis, as almost all matches were played behind closed doors. After removing the first two rounds of each season, our final dataset consists of 775 observations, with match-level data collected from www.football-lineups.com and www.soccerdonna.de. As our dependent variable, attendance, proved to be right-skewed, we take the natural logarithm.

The dataset includes a rich set of variables to control for the unobserved heterogeneity of the home and away teams, along with the season, day, and month the match takes place in. Specifically, home, and away team dummies account for the fixed differences across teams, such as populations and team importance. We further appraise the importance of controlling for home city and only home team fixed effects, which are included as additional specification within the appendix. Season specific fixed effects are included to capture the trend evolution in attendance. This treatment of seasons also allows for the uncovering of non-linear trends in attendance and allows us to assess the effect of international competitions on domestic attendance. Note, this inclusion of season specific effects will also control for unobserved country level macroeconomics variables such as inflation and unemployment, along with capturing the effect of GDHI per Head. Finally, month and day specific fixed effects allow us to control for the unobserved influence of seasonal components on attendance.

We model attendance using a range of variables which are often employed within the literature on attendance demand for men's and women's football. Broadly, we categorise the general determinants of stadium demand following the categorises outlined by Borland and MacDonald (2003). To capture the economic determinants that may affect attendance at women's football, we employ the gross disposable income per head (*GDHI Per Head*), from the home team's ITL3 region, with data for this variable collected from the Office for National Statistics (2022). To account for unreleased datapoints, specifically values for 2022 and 2023, the average growth rate of the region is used to estimate the missing figures. Like attendance, this variable is logarithmised. We further include the number of years each team has played in the WSL to assess the importance of the team's foothold within the market (*Seasons Played in WSL [H/A]*).

Like previous papers in the literature studying attendance demand in women's football, we do not include ticket price among the set of covariates; see, for example, Meier et al. (2016) and Valenti et al. (2020). It is noted that clubs within the WSL only charge a small fee for admission (or even allow for free entrance), with the average women's ticket price for the 2023/24 season approximately £8.71 (calculated using prices collected from each team's website), 79% lower than that of the average men's football ticket price in the same season. However, attendance is also affected by a myriad of indirect costs. For example, the distance, measured in kilometres, between the competing teams' home stadiums (collected from www.distancefromto.net) is included as a proxy for the travelling costs faced by away fans. Finally, to account for the opportunity cost of watching the women's team over the men's team, a pair of dummy variables are employed for when the women's team is playing on the same day the associated men's team is playing at home or away (*Men's Team Playing at Home, Men's Team Playing Away*).

The viewing quality of the football match is operationalised using several variables. First, the inclusion of day and month fixed effects accounts for the influence of match timing of attendance; however, we expand on this by following Meier et al. (2016) with the inclusion of a dummy variable indicating matches played after 6pm (*Evening*). To capture the effect of adverse weather on match attendance, the average temperature in Celsius and amount of rain in millimetres for the day of the match in observation are included.[4]

Like GDHI per Head and Distance, we use the natural logarithm of each variable, with the data for each collected from visualcrossing.com. Finally, like Chahardovali et al. (2023), Meier et al. (2016), and Valenti et al. (2024), we appraise the importance of stadiums on match attendance. We expand on previous approaches by including two dummy variables. First, to account for the influence of improved stadium capacity and quality, along capturing the effect of increased ticket availability and affordability for fans who want to attend matches at the stadium, but struggle to obtain men's team tickets (Valenti et al., 2024), we include a variable that indicates matches played in the men's main stadium (*Men's Home Stadium*). Expanding from this, our second dummy indicates matches that are specially relocated to the men's main stadium (*Special Relocation*). We include this variable to capture the previously stated effects of playing in the men's stadium, along with any changes in demand due to increased excitement. Such matches also tend to be heavily marketed by

[4]We may expect that the day and month fixed effects capture the effect of temperature and match scheduling on attendance, however, we include such variables to understand if they are statistically significant even with the inclusion of fixed effects.

the football clubs, with the social media pages for the men's team or prominent players from the men's side advertising such matches.[5]

A range of covariates are employed to capture the specific characteristics of the contest. The current form of the home and away teams is operationalised with the league position of each team before the match (*League Position [H/A]*). Similarly, the medium-term quality of the teams is captured with the league position of each team in the prior season (*Position in Previous Season [H/A]*). Finally, we control for the long-term quality and potential branding effects of the competing teams by including the number of trophies won by each team up to the match in observation (*No. Trophies Won [H/A]*). Further characteristics of the contest are operationalised by a set of dummy variables. To uncover a potential momentum effect on attendance, we follow Buraimo, Tena, and De La Piedra (2018) with the inclusion of variables indicating if the home team won or drew their previous match (*Win/Draw Previous*). Further, we include a variable indicating if the home team won a trophy in the previous season (*Trophy in Previous Season*). Finally, similar to Wooten (2018) reviewing men's football and Valenti et al. (2020) reviewing women's, we appraise the effect that rivalries between clubs has on attendance by including an indicator for matches classified as derbies (*Derby*).[6]

Given the focus of this research, we employ a similar set of variables mentioned in the previous paragraph for the men's teams to understand any potential spillover effects. Thus, we include the number of major trophies won by the home and away teams (*Trophies Won by Men's Team [H/A]*), a dummy variable taking the value one if the home men's team won a trophy the previous season (*Men's Team Trophy in Previous Season*), along with the male clubs' positions the previous season (*Men's Team Position in Previous Season [H/A]*).

Like much of the attendance demand literature, we appraise the effect of competitive balance on stadium attendance. Uniquely, the presence of women's team integration provides interesting implications for competitive balance. As noted, integrated football teams benefit from greater financial resources and enhanced footballing facilities, which can widen the competitive gap between teams associated with large football clubs and those associated with smaller clubs, thereby reducing competitive balance in matches between them. Figure 2 provides suggestive evidence of this disparity in competitive balance, where we see matches played between traditional 'big-6' and non- 'big-6' sides showing lower match outcome uncertainty (as measured by the Theil index). We can also see this gap in match uncertainty widening from 2014, likely due to the salary cap, which benefitted competitive balance by allowing teams with smaller financial resources to compete for talent, being relaxed at the time (Wrack, 2019).

[5]As an example, the Liverpool FC social media pages released a video of Virgil van Dijk, the men's team captain, advertising the match between Liverpool and Manchester City on the 13th October 2024 which was played at Anfield, the main ground of Liverpool FC (@liverpoolfcw, 2024).

[6]The matches we define as historical derbies within the dataset are Arsenal vs Tottenham, Aston Villa vs Birmingham City, Everton vs Liverpool, Liverpool vs Manchester Utd., and Manchester City vs Manchester Utd.

Figure 2: Dynamics of Outcome Uncertainty (Theil Index)

Note: The teams classified as 'big-6' are Arsenal, Chelsea, Liverpool, Manchester City, Manchester United, and Tottenham Hotspur; A larger value for the Theil index represents more uncertain matches.

To review the effect of competitive balance on stadium attendance, the standard procedure for operationalising outcome uncertainty is followed, with betting odds used to compute the probabilities of each result. Using these probabilities, the match outcome uncertainty (Uncertainty of Outcome) is calculated using the Theil index (Theil, 1967):

$$Theil = \sum_{i=1}^{3} \frac{p_i}{\sum_{i=1}^{3} p_i} log(\frac{\sum_{i=1}^{3} p_i}{p_i}) \tag{1}$$

where p_i represents the probability of a home win, draw, or loss in match i. Betting odds were collected from oddsportal.com and aiscore.com. To account for over-round, each probability for match outcome j was calculated as such:

$$\frac{ImpliedOdds_j}{\sum_{j=1}^{3} ImpliedOdds_j} \tag{2}$$

For matches that did not have betting data available, we adopted a Poisson Distribution model to fill the remaining matches.[7] It is noted that this is a rough, but ultimately appropriate, proxy for match probabilities, with 31 observations needing computation. To account for the potential asymmetric relationship between outcome uncertainty and attendance, and to capture the effect of reference dependent preferences, the home team's win

[7]The model we use follows Chalikias, Kossieri, and Lalou (2020). We compute the probability of the match result up to 8-8 and sum the probabilities of the home team winning, drawing, or losing to get the estimated probabilities of each outcome.

Table 1: Summary Statistics

	Mean	Std. Dev	Min	Max
Attendance	1807.93	4116.42	95	46881
Distance	173.21	97.38	1.06	449.01
Temperature	10.74	4.52	-3.2	22.1
Rain	1.82	3.6	0	23.16
GDHI per Head	23528.8	13440.2	12115	66198.5
League Position (H)	5.73	3.1	1	12
League Position (A)	5.77	3.03	1	12
Position in Previous Season (H)	6.1	4.1	1	27
Position in Previous Season (A)	6.17	4.15	1	27
No. of Trophies Won (H)	4.53	9.32	0	35
No. of Trophies Won (A)	4.8	9.81	0	35
Trophy Won in Previous Season (H)	0.19	0.39	0	1
Win in Previous Game	0.4	0.49	0	1
Draw in Previous Game	0.17	0.38	0	1
Derby Match	0.03	0.18	0	1
Special Relocation	0.07	0.26	0	1
Men's Home Stadium	0.09	0.28	0	1
Evening	0.32	0.47	0	1
Seasons Played in WSL (H)	5.22	3.34	1	13
Seasons Played in WSL (A)	5.27	3.38	1	13
Probability of Winning	0.43	0.27	0.01	0.96
Uncertainty of Outcome (Theil Index)	0.85	0.23	0.18	1.09
Trophies Won by Men's Team (H)	14.53	13.98	0	45
Trophies Won by Men's Team (A)	15	14.27	0	45
Men's Team Trophy in Previous Season (H)	0.2	0.4	0	1
Men's Team Playing at Home	0.06	0.25	0	1
Men's Team Playing Away	0.1	0.3	0	1
Men's Team Position in Previous Season (H)	18.72	21.84	1	109
Men's Team Position in Previous Season (A)	19.24	23.34	1	109

probability is also included within the model (*Probability of Winning*).[8] Summary statistics for each of the variables are presented within Table 1.

5 Empirical Analysis

Our empirical strategy follows previous attendance studies using match-level data. Our preferred specification takes the following form:

$$Ln(Attendance_{g,t}) = \beta_0 + \beta_1 X_{g,t} + \tau_t + \Omega_d + \Omega_m + \pi_h + \phi_a + \varepsilon_{g,t} \quad (3)$$

where the dependent variable is the natural log of attendance for a fixture g in season t.

[8]See Buraimo et al. (2018) who adopt a similar approach.

$X_{g,t}$ is a vector of covariates described in the previous section. τ_t denotes season fixed effects, and Ω_d and Ω_m are dummy variables indicating the day of the week and month when the match is played. The terms π_h and ϕ_a are home and away team fixed effects. Finally, $\varepsilon_{g,t}$ represents the stochastic error term. As capacity constraints are not an issue for teams within the FA WSL, it is not necessary to adjust our specification accordingly.

Table 2 presents the estimation results of the model outlined in equation 3. We show the model specifications with and without home and away team fixed effects to appraise the effect of team-specific heterogeneity on the model results. [9]

It is notable that the coefficient sizes and significance of most variables are consistent between the models with and without team fixed effects. However, we see that the position in the previous season for the competing teams and the number of trophies won by the away men's and women's teams become insignificant predictors when we include the fixed effects. Similarly, the effect of outcome uncertainty and scheduling conflicts when the men's team are playing at home lose their significance. We find that, consistent with our expectations, current performance indicators are significant predictors of attendance, especially for the home team. In our preferred specification, a one-place movement up the current league table by the home and away team is concomitant with a 6% and 3.3% increase in attendance respectively. This is consistent with the findings by Meier et al. (2016), Stephenson (2024), and Valenti et al. (2024) in women's football and Buraimo (2014) and Forrest and Simmons (2006) in English men's football, though the measure of current form is slightly different. Despite this, we do not find that short or long-term measures of a team's quality are statistically significant in our preferred model. Similarly, the number of years that the home team has played in the WSL is significant, with each year additional year played in the top league being associated with a 12.6% attendance increase. Indicators of historical performance for the home team are also significant predictors of attendance, regardless of model specification, such that a better position for the home team in the season prior leads to an expected improvement in attendance. The momentum effect from winning or drawing the previous match is insignificant across all model specifications.

Derby matches significantly impact attendance across all model specifications, like previous findings in the literature on men's football (Wooten, 2018). Given it is a binary dummy variable, the coefficient must be adjusted to indicate the percentage increase in attendance (Halvorsen & Palmquist, 1980). After the adjustments, calculated as $100 * exp(coefficient)–1$, derby matches increased expected attendance by 102.4% in our preferred specification.

We find that the amount of rainfall during a matchday is a significant predictor of match attendance, with a 1% increase in the amount of rainfall leading to an expected 5.2% decrease in attendance. Again, this result is consistent with previous studies on the demand for

[9] We present additional specifications with city and home team fixed effects in Table A within the appendix; however, we do not find notable differences between model specifications compared to our preferred model.

Table 2: Determinants of Attendance Demand in the WSL

	(1)		(2)	
	Estimate	Std. Error	Estimate	Std. Dev
Ln Distance	-0.083***	(0.022)	-0.085***	(0.018)
Ln Temperature	0.132	(0.097)	0.123	(0.092)
Ln Rain	-0.058***	(0.021)	-0.052***	(0.018)
Ln GDHI per Head	0.027	(0.079)	-1.162	(1.191)
League Position (H)	-0.076***	(0.018)	-0.060***	(0.014)
League Position (A)	-0.049***	(0.010)	-0.033***	(0.011)
Position in Previous Season (H)	-0.017*	(0.010)	-0.011	(0.010)
Position in Previous Season (A)	-0.011*	(0.007)	-0.008	(0.007)
No. of Trophies Won (H)	0.006	(0.004)	-0.038	(0.034)
No. of Trophies Won (A)	0.004**	(0.002)	0.021	(0.017)
Trophy Won in Previous Season (H)	-0.038	(0.090)	-0.061	(0.094)
Win in Previous Game	0.036	(0.042)	-0.013	(0.044)
Draw in Previous Game	-0.014	(0.049)	-0.018	(0.046)
Derby Match	0.697***	(0.188)	0.705***	(0.173)
Special Relocation	0.760***	(0.231)	0.981***	(0.210)
Men's Home Stadium	0.422***	(0.114)	0.386**	(0.163)
Evening	0.015	(0.065)	-0.074	(0.057)
Seasons Played in WSL (H)	-0.047***	(0.015)	0.126*	(0.074)
Seasons Played in WSL (A)	0.000	(0.007)	-0.007	(0.052)
Probability of Winning	0.627***	(0.200)	0.754***	(0.202)
Uncertainty of Outcome (Theil Index)	-0.222*	(0.117)	-0.040	(0.099)
Trophies Won by Men's Team (H)	0.000	(0.003)	0.099**	(0.041)
Trophies Won by Men's Team (A)	0.009***	(0.002)	-0.031	(0.022)
Men's Team Trophy in Previous Season (H)	0.292***	(0.066)	0.126*	(0.074)
Men's Team Playing at Home	-0.107*	(0.063)	-0.036	(0.055)
Men's Team Playing Away	-0.133**	(0.068)	-0.131**	(0.059)
Men's Team Position in Previous Season (H)	-0.002	(0.003)	0.003	(0.005)
Men's Team Position in Previous Season (A)	0.000	(0.001)	0.002	(0.003)
Home Team Fixed Effects	No		Yes	
Away Team Fixed Effects	No		Yes	
Adjusted R^2	0.694		0.761	
N	775			
Dependent Variable Mean	Absolute value - 1807.93 Logged value - 7.50			

Note: *p<0.1; **p<0.05; ***p<0.01; Standard errors in parentheses, clustered at the home team and season levels. Dependent variable: Log(*Attendance*).

women's football, such as Meier et al. (2016) and Valenti et al. (2024). However, unlike previous studies, we do not find temperature is a significant predictor of attendance, though this is likely because the effect is captured by the month fixed effects. Similarly, we find matches played in the evening has no significant effect on attendance. Though, evening matches are also correlated with the day the match is played, with weekday matches more likely to be played during the evening. Due to this, the day-specific fixed effects already capture the effect of match timing.

Turning to the effect of indirect costs, such as opportunity or travelling costs, our results also show the significant impact of distance between teams, with a 1% increase in distance decreasing attendance by 8.5%. Although we do not directly observe attendance by visiting fans, we suggest that this is due to the increased travelling costs faced by supporters lowering demand. Similarly, it may capture some of the effect of increased distances lowering the likelihood of teams having a rivalry. Likewise, in our preferred specification, matches with scheduling conflicts with that of the men's team do not significantly influence WSL attendance, although conflicts with matches where the men's team are playing away is negative and weakly significant.

Finally, like the results presented by Valenti et al. (2020), the increased probability of a home win is a significant indicator of attendance across all model specifications. However, when controlling for the heterogeneity of the competing teams, no support is found in favour of a positive impact from match outcome uncertainty, suggesting fans of English women's football are unconcerned about competitive imbalances between teams, and adding to the literature that does not support the outcome uncertainty hypothesis (Baydina et al., 2021; Valenti et al., 2020).

Male Team Spillover Effects

We turn our attention to the study of potential effects that related male teams have on the women's team. With this analysis, it is essential to emphasise that the econometric framework already controls for the home and away team heterogeneity and the performance of the female team. Therefore, the coefficients associated with the performance of the men's team could be deemed to be directly attributable to the men's team (the other product of the company) rather than just the club's brand name. Table 2 shows that attendance is significantly affected at the conventional levels by the number of trophies the male home team has won, and if the men's home team won a trophy the previous season. Specifically, attendance increases by an estimated 9.9% for each trophy won by the men's team, with attendance seeing a further expected increase of 13.4% after the men's team won a trophy.

Similarly, we find matches played in the men's stadium see a significant increase in demand compared to other stadiums. Matches that are played in the men's stadium, but are not specially relocated, see an expected increase in attendance of 47.1%. We suggest that this is due to the increased quality and capacity of these stadiums making matches in these stadiums more attractive. Similarly, this result may also capture the demand effect of stadium proximity to the teams city centre on demand, like the findings of Chahardovali et al. (2023), given the mean distance of a men's stadium from the city centre within our context is 6.3 km, compared to 14.6 km otherwise. Like the prior result, specially relocated matches see a statistically significant increase in attendance of 167%. We similarly suggest this result captures the same influence on attendance as the previous covariate but is larger because of the increased marketing and excitement that these matches offer to fans who do not usually go to women's matches.

International Competition Spillover Effects

Following several studies that examine attendance demand for women's football (see LeFeuvre et al. (2013) and Stephenson (2024)), we evaluate the demand spillover from international competitions on domestic league attendance. Note that the estimated parameters associated with the season specific dummies in the models are not informative for such a purpose as they can only be interpreted in comparison with a reference season (the 2011 season). Instead, to test the effect, we use the estimates of our preferred specification to assess whether the increase in attendance from the season before an international competition to the season after is statistically different from the increase across the two prior consecutive seasons. We do this using F-statistics to test the null hypothesis. For example, considering the 2019 World Cup, this hypothesis can be formalised as:

$$H_0 : (\beta_{2019-20} - \beta_{2018-19}) - (\beta_{2018-19} - \beta_{2017-18}) = 0 \tag{4}$$

We further present the results of F-tests that review if attendance from the season after an international competition is statistically different to the season prior. Though this doesn't necessarily capture the effect of international competitions on domestic attendance, it is still interesting to see if the attendance increase from the prior seasons are significant.[10]

We present the results of such tests in Table 3, where the difference in coefficients column presents the increase from only the prior season and the right hand column shows the difference in coefficient differences from consecutive seasons, which we consider as the demand spillover effect of an international competition.[11] We find interestingly that the significance of the spillover effect from international competitions is not consistent across competitions, instead seemingly related to the success of the English women's national

[10]Note, some international competitions in our analysis conclude mid-WSL season. To account for this, we compare attendance increases from the competitions season to the following season, relative to the prior seasons increase. For example, for the 2013 World Cup, we compare the attendance change from the 2013 to 2014 seasons to the change from 2012 to 2013. Similarly, to address the Covid-19 disruptions, for the 2022 Euros, we compare the increase from 2021/22 to 2022/23 with the change from 2019/20 to 2021/22.

[11]We also provide the relevant F-tests for all alternative model specifications within Table B in the appendix. However, we find the results do not differ much across specifications.

Table 3: F-Test for Differences in Season Specific Effects

	(1)		(2)	
	Estimate	Std. Error	Estimate	Std. Dev
2013 Euros (Group-stage)	2.446	0.12	0.567	0.45
2015 World Cup (3rd Place)	0.009	0.92	5.182**	0.02
2017 Euros (Quarter finals)	3.668*	0.06	0.005	0.94
2019 World Cup (4th Place)	3.541*	0.06	1.045	0.31
2022 Euros (Winners)	25.030***	0.00	33.471***	0.00

Note: *p<0.1; **p<0.05; ***p<0.01. Table shows the F-test on the coefficient estimates collected from specification 2 in Table 2. Column (1) presents an F-test on the difference in estimates, column (2) presents an F-test on the differences in estimates differences.

team. Reviewing the difference in coefficient differences, we see that the 2015 World Cup, where the English national team placed third, and the 2022 European Championships, where the national team won the tournament, had significant spillovers on the demand for women's football, with the 2022 Euros having a particularly strong impact. This result supports the strategy of the FA in focusing on the strength and success of the English women's team to increase demand for domestic competitions and suggests future tournaments may affect spectatorship in the same way, especially in the case of future international success by the English women's national team.

6 Robustness Analysis

In this section, we test the sensibility of our results to three different considerations. First, some papers control for the persistence of consumption habits in attendance demand models by including attendance in the previous match of the home team in its stadium among the covariates. For example, Buraimo et al. (2018) and Meier et al. (2016) find a significant inertial component in attendance at men's and women's football, respectively. With this variable, it should be noted that including attendance in the previous match in the set of covariates affects the interpretation of the other model coefficients. This is because any variable effect on current attendance may indirectly affect future attendance levels through the effect on the lagged dependent variable. Due to this phenomenon, we follow Dobson and Goddard (2001) and compute steady-state coefficients for the set of covariates. Thus, we augment our preferred specification in the previous section by including home attendance in the previous match. We show estimation results for the dynamic and static specifications in Models 1 and 2, respectively.

The second robustness test involves employing indicators of match significance. A discussion of the use of such variables within previous literature can be found in Buraimo, Forrest, McHale, and Tena (2022). Jennett (1984) was the first paper employing proxies of match significance, in his case to explain Scottish football attendances. However, he proposed an ex-post measure based on the final points total at the end of the season. Other approaches to estimate match significance focus on the distance the competing team is from a final point target, which does not explicitly account for the expectations of future match outcomes. For example, in Goddard and Asimakopoulos (2004) and Buraimo and Simmons (2015), a match is significant if a club can still win the title by securing victories in all remaining fixtures while all other clubs only get one point per game. Similarly, Forrest, Simmons, and Buraimo (2005) regard a match as significant if at least one of the clubs is close to the relevant positions in the table. However, such variables did not play an important role in explaining television audiences. As such, Tainsky and Winfree (2010) and Buraimo et al. (2022) employ an index of match significance based on forecasting future outcomes to explain baseball stadium attendance and TV audiences for the English Premier League, respectively. Thus, we follow Buraimo et al. (2022) in the estimation of match significance based on the difference between the final league positions if the home or away team win the upcoming match, compared to losing it. To compute the final league table, they simulate match results using match forecasting models. Following similar arguments, we estimate the match significance for winning the WSL championship, qualifying for European competitions, and relegation in each game using the expected probabilities for future matches to compute league standings.[12] We include such variables in our preferred specification to examine how this consideration affects estimation results in Model 3.

Our final robustness analysis substitutes home and away dummies with fixture-specific dummies included in Model 4. Whilst fixture dummies are not usually employed in the attendance demand literature, due to the important reduction they impose on the number of degrees of freedom, it is plausible to assume that these variables may capture many unobserved match attendance predictors, such as match significance or team rivalries.

Table 4 reports the results of these experiments. Importantly, the main results concerning the importance of a specific group of match characteristics and male team spillover effects are not affected by the model augmentations. More specifically, performance indicators for the women's teams such as the current home and away positions, along with the probability of the home team winning the coming match are significant across all model specifications. Moreover, in all models, at least one indicator of indirect costs (Distance, Temperature, and Rain) significantly predicts attendance. Regarding spillover effects, the number of trophies won by the home men's team and special relocation matches are significant across all model specifications.

Focusing on the specific components of Models 1 and 2, we see habit persistence is a significant component in WSL attendance. It is notable that this effect is relatively small (0.110),

[12] Although not incorrect, a simulation of the league is unnecessary. While in knock-out tournaments, fixtures depend on previous match results (and each simulation provides a different set of matches), fixtures are fixed in league competitions. As such, it is possible to use the probabilities of different match outcomes independently rather than approximate them using simulations.

which makes the short- and long-term coefficients between each model relatively similar. When viewing Model 3, match significance indicators do not play a main role in explaining attendance apart from when the home team is in contention for relegation (which has a negative effect) and the away team is in contention for the championship. At first glance, this may seem in contradiction with Buraimo et al. (2022), who find that match significance is an important predictor of English Premier League TV audiences. However, this could be explained as attendance in female football could be more insensitive to match significance than watching men's football on television. Finally, despite such a large number of additional variables (267), including fixture dummies, does not have a qualitative impact on coefficients in the model, apart from the derby variable being dropped due to multi-collinearity.

As a final robustness check, we one-by-one remove a home side from our preferred specification to review any potential changes in our estimates on variables relating to potential male team spillover effects and appraise if the results are influenced by one specific club. We present the results of such estimation within Figure A in the appendix. The results show that the size of the estimates do not change significantly when we remove any team; however, it is notable that there is a decrease in the effect size of most covariates when we remove Chelsea from our estimation, along with a large decrease in the effect size of playing in the men's home stadium when we remove Reading, though the sign of the coefficient estimates does not change in any new estimation.

7 Discussion Remarks and Future Research

The present paper studies the determinants of stadium attendance in WSL games from 2011 to 2022/23. We found that team performance, the probability of the home team winning the match, derbies, and indirect costs, such as the distance between the home and the away teams, are essential determinants of stadium attendance. The analysis also enhances the importance of controlling for unobserved home and away team heterogeneity, as some variables are no longer significant or change their signs once we include these controls. This is the case, for example, with trophies won in previous years.

Overall, the results of this paper are like previous analyses of men's sports. Many of these papers also find that performance (Dobson & Goddard, 2001), derbies (Wooten, 2018), and home win probabilities (Buraimo et al., 2018) are significant predictors of stadium attendance. Our study also shows that, consistent with previous analyses for men's football, outcome uncertainty is not an important predictor of attendance (Brandes Franck, 2007). Instead, fans who are predominantly home team supporters are more attracted by the probability of home win (Buraimo Simmons, 2008). Another finding from our analysis is the spillover effect of the men's team on attendance at women's matches. In particular, the previous performance of the home men's teams significantly affected attendance even after controlling for home and away club dummies, indicating a spillover effect from men's to women's teams. Accordingly, most Premier League and WSL teams could be conceived as part of multiproduct companies that share a proportion of their fans. This result has clear implications for the economic analysis of the club, as investment returns could be different if the spillover effects are considered.

Finally, our study highlights the importance of the international success of the women national team to boost league attendance. More specifically, compared to any of the previous years, the victory of the English women's team in the UEFA Women's Championship had an exceptional effect in boosting attendance. This result aligns with Meier et al. (2016), who also found an important impact from winning the Women's World Cup in 2011 for women's league soccer in Germany.

Such results provide a range of implications for the stakeholders of women's football teams. Most predominantly, we show that women's football teams can benefit through the association to larger football clubs usually associated with a men's football team, both through a brand spillover from the success of the men's side, but also by facilities sharing such as sharing the football club's main stadium (like the finding by Hadwiger et al. (2024)). This provides some support for the brand strategy employed by the FA which likely assisted the women's teams in their early stages of growth. Out findings also suggest that the FA and other stakeholders need to carefully consider the scheduling decisions of the league such that football clubs can see the maximum benefit. Specifically, scheduling derby matches in times where the club's main stadium is free, such as during international breaks, can see a large benefit to demand; similarly, matches should be scheduled to avoid conflict with the men's team playing.

Though we provide an extensive analysis on the determinants of women's football demand, there are certain limitations. Firstly, the present analysis focuses exclusively on stadium attendance. Although this is a good measure of demand, it does not consider the other avenues that football supporters consume women's football, such as through social media or, importantly, sports broadcasting. Given the WSL has recently seen major investment through TV-deals, understanding how demand for broadcasted women's football is determined is an important avenue for future research. Similarly, given the availability of the data, we do not consider issues with the dependent variable such as no-show behaviour or the share of season tickets. This presents potential issues with the bias of estimates, as Popp, Simmons, Shapiro, and Watanabe (2023) show. Similarly, Schreyer and Torgler (2021) find that the number of no-shows is associated with the amount of distributed season tickets. Again, with the growing availability of data for women's football, this is something future research should consider.

We hope this paper incentivises future research on factors that enhance consumer interest in women's football. We can mention at least two potential research avenues. The first one should explore whether a similar effect can be uncovered for television audiences as spectators are a different type of consumer potentially interested in other characteristics of the game; see Buraimo et al. (2020) and Buraimo et al. (2022) for examples of this type of analysis in men's football. In the case of female football is interesting to study both the factors that drive television audience and how television can stimulate attendance across WSL. A second line of research concerns the size of gains to sports clubs from widening their product portfolio. For example, here we have considered whether co-branding between men's and women's football impacts the popularity of the add-on product. Many Greek and Spanish football clubs have teams competing not only in female as well as male competitions but also across sports, for example in basketball and handball as well as football. Moreover, in England leading clubs have expanded into e-sports. Our paper suggests that an adequate analysis should consider the presence of spillovers across the different teams of a professional club.

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